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#include<stdio.h>
#include<stdlib.h>
#include<pthread.h>
#include<semaphore.h>

#define SIZE 3

int buffer[SIZE];
int in = 0, out = 0;

sem_t empty, full, mutex;

void *producer(void *arg){
    int item;
    sem_wait(&empty);
    sem_wait(&mutex);
    item = rand() % 100;
    buffer[in] = item;
    printf("Producer produces the item %d\n", item);

    in = (in + 1) % SIZE;

    sem_post(&mutex);
    sem_post(&full);

    pthread_exit(NULL);
}

void *consumer(void *arg){
    int item;
    sem_wait(&full);
    sem_wait(&mutex);

    item = buffer[out];
    printf("consumer consumes item %d\n", item);

    out = (out + 1) % SIZE;

    -- INSERT --

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    item = buffer[out];
    printf("consumer consumes item %d\n", item);

    out = (out + 1) % SIZE;

    sem_post(&mutex);
    sem_post(&empty);

    pthread_exit(NULL);
}

int main(){
    int choice;
    pthread_t p, c;

    sem_init(&empty, 0 , SIZE);
    sem_init(&full, 0 , 0);
    sem_init(&mutex, 0 , 1);

    while(1){
        printf("\n 1.producer\n2.consumer\n3.exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch(choice){
            case 1:
                if(sem_trywait(&empty) == 0){
                    sem_post(&empty); //restore
                    pthread_create(&p, NULL, producer, NULL);
                    pthread_join(p, NULL);
                }else{
                    printf("Buffer is full!!\n");
                }
                break;

            case 2:
                if(sem_trywait(&full) == 0){
                    sem_post(&full); // restore
                    -- INSERT --

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sem_init(&full, 0, 0);
sem_init(&mutex, 0, 1);

while(1){
printf("\n 1.producer\n2.consumer\n3.exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch(choice){
case 1:
if(sem_trywait(&empty) == 0){
sem_post(&empty); //restore
pthread_create(&p, NULL, producer, NULL);
pthread_join(p, NULL);
}else{
printf("Buffer is full!!\n");
}
break;
case 2:
if(sem_trywait(&full) == 0){
sem_post(&full); // restore
pthread_create(&c, NULL, consumer, NULL);
pthread_join(c, NULL);
}else{
printf("Buffer is empty!!\n");
}
break;
case 3:
exit(0)
default:
printf("Invalid choice\n");
}
}
return 0;
}
}
-- INSERT --

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sadha237@localhost:~ -- ./pc

[sadha237@localhost ~]$ vi producer_consumer.c
[sadha237@localhost ~]$ gcc producer_consumer.c -o pc -lpthread
[sadha237@localhost ~]$ ./pc

1.producer
2.consumer
3.exit
Enter your choice: 1
Producer produces the item 83

1.producer
2.consumer
3.exit
Enter your choice: 2
consumer consumes item 83

1.producer
2.consumer
3.exit
Enter your choice: 2
Buffer is empty!!

1.producer

```

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sadha237@localhost:~ — ./pc

1.producer
2.consumer
3.exit
Enter your choice: 1
Producer produces the item 86

1.producer
2.consumer
3.exit
Enter your choice: 1
Producer produces the item 77

1.producer
2.consumer
3.exit
Enter your choice: 1
Producer produces the item 15

1.producer
2.consumer
3.exit
Enter your choice: 
```